Khushali Samderiya

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Software Engineer with experience building scalable, reliable backend cloud services, including fault-tolerant APIs and distributed workflows. Strong computer science foundation with applied AI experience, seeking Software Engineering roles.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, NoSQL, HTML5/CSS3
Backend & Web: React.js, Node.js, Flask, FastAPI, RESTful APIs
Cloud & Infra: AWS (Lambda, API Gateway, S3, CloudFront), GCP, Docker, Kubernetes, Jenkins, Git, CI/CD
Data & ML: MySQL, PostgreSQL, Redis, JDBC, TensorFlow, PyTorch, Scikit-learn, Hugging Face (LLMs), Keras, OpenCV

Work Experience

AI Software Engineer | Living Stones Foundation *Aug 2025 – Aug 2026*

- Led backend architecture for the organization's payment system, improving payment reliability by 45% by building a PostgreSQL-backed transaction state machine with anomaly checks for stuck payments and failed webhooks.
- Reduced duplicate transactions by 75% by implementing idempotent payment APIs and authenticated webhook handlers, coordinating with frontend and data teams on retry and event handling.
- Reduced payment API latency by 30% by optimizing transaction queries and moving log classification, retry analysis, and reconciliation checks to asynchronous worker pipelines
- Improved payment debugging efficiency by building a diagnostics pipeline that classified webhook failures by error code, transaction state, retry count, and gateway response to surface root-cause patterns.

Graduate Research Assistant | Bioinformatics Lab, University of Colorado *June 2024 – May 2025*

- Improved classification precision by 9% by building CNN pipelines with residual and attention-based architectures.
- Increased model accuracy by 12% through biological dataset augmentation and stratified K-fold validation.
- Reduced preprocessing runtime by 40% by reusing transformed datasets and eliminating redundant computation.

Software Engineer | Independent Consultant *Feb 2022 – July 2023*

- **Midaz Technocraft – Internal Operations Platform**
  - Built a full-stack internal operations platform (JSP/Java, MySQL), supporting ~**$120K** annual cost savings, by automating inventory, order, and cash-flow workflows.
  - Reduced average API response time by ~**40%** by adding Redis caching for frequently accessed data and using database transactions to prevent partial order update failures.
  - Improved concurrent request handling by 30% by optimizing API endpoints, offloading non-critical work to an async task queue, and using thread pools with synchronization for thread-safe access
- **Retail Startup – Serverless E-commerce Checkout**
  - Reduced cart abandonment from **45% to 15%** by building a serverless checkout using AWS Lambda, API Gateway, PostgreSQL, and CloudFront.
  - Improved release reliability through unit and integration tests covering API responses, payment failures, and order-state transitions.
- **Apparel Startup – Payments & Order Lifecycle**
  - Reduced payment failures by **80%** by rebuilding checkout workflows using AWS Lambda, React and API's
  - Enabled real-time order tracking and automated refunds through a webhook-driven order state machine (initiated, pending, confirmed, failed, refunded)

Projects

TriModel Financial Analyzer | University of Colorado *Jan 2024 – Dec 2024*

- Improved stock trend prediction accuracy to **84%** by training time-series, technical, and BERT-based sentiment models.
- Built scalable data pipelines to collect, clean and align price data, indicators, and financial news for model training.
- Reduced training time by **30%** by restructuring training workflows to reuse preprocessing outputs across experiments.

Real-Time Trading Analytics Platform | J.P. Morgan Chase *Sep 2021 – Oct 2021*

- Reduced manual analysis time by **35%** by developing real-time stock dashboards using WebSockets and RESTful APIs.
- Reduced system latency by **40%** by optimizing data transformations through batching and distributed processing.
- Improved system responsiveness by **25%** by refactoring Java services for concurrent request handling.

Education

University of Colorado, MS in Computer Science, CGPA: 3.8 *Aug 2023 - May 2025*

Courses: Data Structures and Algorithms, AI, NLP, Advanced Software Engineering, Software Design

St. Francis College for Women, BS Hons in Software Engineering, CGPA: 3.67 *Jun 2018 - Dec 2021*

Courses: Java, C, C++, Database Systems, Object Oriented Programming, Operating Systems, Software Architecture